66 Type-IV collinear spin layer groups

Néel directions, SOC magnetic layer groups and layer-resolved intrinsic nonlinear Hall constraints

Coverage. All 66 entries of Table S6, in the original order, expanded into **510 direction branches**: 8 groups have 3 branches, 48 have 7, and 10 have 15. Directions related by a parent symmetry are retained in the fixed source coordinate setting so that the displayed tensor indices stay meaningful. These 510 rows are direction strata, not 510 distinct MLG types.

Source. J.-X. Du, S. Zeng and Y.-J. Zhao, *Stacking theory for bilayer two-dimensional magnets*, Phys. Rev. B **113**, 214455 (2026), DOI: 10.1103/g471-xrkl; Supplemental Material, Table S6, pp. 84–95. The source supplies the cSLGs and operations. The Néel-direction branches, SOC MLGs and response constraints in this report are calculated results.

Direction notation. $\mathrm{AX}_{[uvw]}$ denotes a special axis; $\mathrm{PL}_{(hkl)}$ denotes a direction in a special plane excluding all axes lying in that plane; GEN denotes a direction outside every listed axis and plane. Only symmetry-distinct stabilizer strata of a given group are introduced; for example, the first group has no special in-plane axis and needs only $\mathrm{AX}_{[001]}$, $\mathrm{PL}_{(001)}$ and GEN.

Coordinates. Tensor components are Cartesian, with z normal to the layer. For rectangular groups, $x \parallel \mathbf{a}$, $y \parallel \mathbf{b}$. For hexagonal groups, $\mathbf{a} = (1, 0, 0)$ and $\mathbf{b} = (-1/2, \sqrt{3}/2, 0)$, while Cartesian $y \parallel \hat{z} \times \mathbf{a}$. Miller indices refer to these crystallographic bases. Thus [010] is not the Cartesian y direction in the hexagonal setting. Hexagonal in-plane operations are displayed with explicit crystallographic directions; $M_{[uvw]}$ has its Cartesian normal along $[uvw]$. The source triplets $\{x, y, xy\}$ and $\{1, 2, 3\}$ correspond as sets to $\{[100], [010], [110]\}$ and $\{[1\bar{1}0], [120], [210]\}$. No individual identification of the source's numbered subscripts is needed.

Operation columns. Translation parts are omitted and repeated point operations are merged. All distinct nontrivial-part SLG point operations are shown on separate lines; the common spin-only factor is implicit. Every MLG row explicitly lists its point operations on one line. Full MLG symbols and numbers are nevertheless identified from the full affine operations in the source, then verified by a basis/origin change preserving the layer normal. An affine witness for every row is included in the data package. Omitting translations from the displayed columns does not redefine these group identifiers.

Tensor convention. Set $A^\ell = \chi^\ell_{xyy}$ and $B^\ell = \chi^\ell_{yxx}$ for $\ell = t, b$, with $\chi_{abc} = -\chi_{bac}$ and in-plane indices. Then $\chi_{xyx} = -B$, $\chi_{yxy} = -A$, and all other in-plane entries vanish. The two-component vector $\mathbf{v}^\ell = (A^\ell, B^\ell)$ transforms as $\mathbf{v}^{g\ell} = q_g R_\parallel \mathbf{v}^\ell$. This statement is specific to the $\mathcal{T}$-odd intrinsic Hall part.

Mark	Constraint on the displayed component	Meaning
$+$	$\chi^t = \chi^b$, allowed	Equal layer contributions
$-$	$\chi^t = -\chi^b$, allowed	Hidden, layer compensated
0	$\chi^t = \chi^b = 0$	Forbidden in each layer
v	No fixed proportional relation between the same component in the two layers	See coupled constraints when starred

A star on a direction indicates additional relations between A and B, listed completely in Appendix A (56 branches). In these rows, v must not be read as absence of all symmetry relations: cross-component relations may remain. All allowed values can accidentally vanish microscopically.

Assumptions and calculation. Fixed structure and collinear order; two physical layers at opposite sides of $z = 0$. The SOC subgroup satisfies $q_g \det(R_g) R_g \hat{\mathbf{n}} = s_g \hat{\mathbf{n}}$, with $q_g = +1$ for g and $q_g = -1$ for $g\mathcal{T}$. Operations with $R_{zz} = -1$ exchange the two layers. Branches are constructed from all real ± 1 eigenspaces and their intersections. These are symmetry-allowed constrained orientations, not predictions of magnetic easy axes or stability.

Source inconsistencies treated explicitly (original strings retained in the package).

S1. 31.5.192 (p. 87), operation 04: change $s_{M_x} = -1$ to $+1$. Both $s_{C_{2y}} = -1$ and $s_{M_z} = -1$ require $s_{M_x} = +1$; the printed SLG symbol also assigns $+1$ to m.

S2. 39.5.260 (p. 89), operation 08: use translation $(0, 1/2, 0)$ for M_x, rather than the printed $(1/2, 0, 0)$. This follows from $\mathcal{P}C_{2x}$ and agrees with parent entry 39.1.256. It affects affine validation, not the displayed point operations or tensor constraints.

S3. 78.5.514 (p. 94): the printed symbol's final ${}^{-1}2$ conflicts with operations 10–12, all of which have $s = +1$. We use $p^{-1}\bar{6}^{-1}m^1 2$, consistent with all 12 listed operations and closure. The original label and numbering are preserved in the raw data.

S4. Lookup entry 77.1.505 (Table S3, p. 32), operation 09: use $s_{M_y} = +1$ instead of the isolated -1, consistent with the FM group and its other operations. This repair is only in the reference catalog used for MLG identification.

Complete table, in Table S6 order

First three columns are merged across all direction branches of one cSLG. Asterisks refer to Appendix A. S1–S3 refer to the source audit on the first page.

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
01 3.3.10 $p^11^11^{-1}2$ SI p. 84	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$	$AX_{[001]}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$PL_{(001)}$	$p\bar{1}12$ 3.1.8	$\{E, C_{2z}\}$	0	0
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
02 4.3.14 $p^11^11^{-1}m$ SI p. 84	Type-IV	$\{1\|E\}$ $\{-1\|M_z\}$	$AX_{[001]}$	$p11m'$ 4.3.14	$\{E, M_z\mathcal{T}\}$	−	−
			$PL_{(001)}$	$p\bar{1}1m$ 4.1.12	$\{E, M_z\}$	+	+
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
03 5.3.19 $p^11^11^{-1}a$ SI p. 84	Type-IV	$\{1\|E\}$ $\{-1\|M_z\}$	$AX_{[001]}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	−	−
			$PL_{(001)}$	$p\bar{1}1a$ 5.1.17	$\{E, M_z\}$	+	+
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
04 6.5.25 $p^11^11^{-1}2/^{-1}m$ SI p. 85	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$	$AX_{[001]}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	−	−
			$PL_{(001)}$	$p\bar{1}12/m$ 6.1.21	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	−	−
05 7.5.32 $p^11^11^{-1}2/^{-1}a$ SI p. 85	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$	$AX_{[001]}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	−	−
			$PL_{(001)}$	$p\bar{1}12/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	−	−
06 19.4.107 $p^12^{-1}2^{-1}2$ SI p. 85	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$	$AX_{[100]}$	$p222$ 19.1.104	$\{E, C_{2z}, C_{2y}, C_{2x}\}$	0	0
			$AX_{[010]}$	$p2'2'2$ 19.3.106	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$AX_{[001]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	−	+
			$PL_{(001)}$	$p\bar{1}12$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$PL_{(010)}$	$p2\bar{1}1$ 8.1.34	$\{E, C_{2y}\}$	−	+
			$PL_{(100)}$	$p2'\bar{1}1$ 8.3.36	$\{E, C_{2x}\mathcal{T}\}$	−	+
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
07 20.3.113 $p^12_1^{-1}2^{-1}2$ SI p. 85	Type-IV	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2z}\}$	$\mathrm{AX}_{[100]}$	$p2_122$ 20.1.111	$\{E, C_{2x}, C_{2y}, C_{2z}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p2_1'2'2$ 20.5.115	$\{E, C_{2x}\mathcal{T}, C_{2y}\mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p2_1'22'$ 20.4.114	$\{E, C_{2x}\mathcal{T}, C_{2y}, C_{2z}\mathcal{T}\}$	−	+
			$\mathrm{PL}_{(001)}$	$p112$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	−	+
			$\mathrm{PL}_{(100)}$	$p2_111$ 9.3.43	$\{E, C_{2x}\mathcal{T}\}$	−	+
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
08 20.4.114 $p^{-1}2_1^12^{-1}2$ SI p. 85	Type-IV	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2z}\}$	$\mathrm{AX}_{[100]}$	$p2_1'2'2$ 20.5.115	$\{E, C_{2x}\mathcal{T}, C_{2y}\mathcal{T}, C_{2z}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p2_122$ 20.1.111	$\{E, C_{2x}, C_{2y}, C_{2z}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p2_12'2'$ 20.3.113	$\{E, C_{2x}, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}\}$	+	−
			$\mathrm{PL}_{(001)}$	$p112$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	+	−
			$\mathrm{PL}_{(100)}$	$p2_111$ 9.1.41	$\{E, C_{2x}\}$	+	−
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
09 21.4.121 $p^12_1^{-1}2_1^{-1}2$ SI p. 85	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p2_12_12$ 21.1.118	$\{E, C_{2z}, C_{2y}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[010]}$	$p2_1'2_1'2$ 21.3.120	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p2_12_1'2'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	−	+
			$\mathrm{PL}_{(001)}$	$p112$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2_111$ 9.1.41	$\{E, C_{2y}\}$	−	+
			$\mathrm{PL}_{(100)}$	$p2_1'11$ 9.3.43	$\{E, C_{2x}\mathcal{T}\}$	−	+
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
10 22.4.125 $c^12^{-1}2^{-1}2$ SI p. 85	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$c222$ 22.1.122	$\{E, C_{2z}, C_{2y}, C_{2x}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c2'2'2$ 22.3.124	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	−	+
			$\mathrm{PL}_{(001)}$	$p112$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c211$ 10.1.45	$\{E, C_{2y}\}$	−	+
			$\mathrm{PL}_{(100)}$	$c2'11$ 10.3.47	$\{E, C_{2x}\mathcal{T}\}$	−	+
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
11 23.4.132 $p^1m^{-1}m^{-1}2$ SI p. 85	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pmm2$ 23.1.129	$\{E, C_{2z}, M_y, M_x\}$	0	0
			$AX_{[010]}$	$pm'm'2$ 23.3.131	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$AX_{[001]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$PL_{(001)}$	$p112$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$PL_{(010)}$	$pm11$ 11.1.50	$\{E, M_y\}$	v	0
			$PL_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
12 24.4.139 $p^{-1}m^1a^{-1}2$ SI p. 85	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pm'a'2$ 24.3.138	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$AX_{[010]}$	$pma2$ 24.1.136	$\{E, C_{2z}, M_y, M_x\}$	0	0
			$AX_{[001]}$	$pma'2'$ 24.5.140	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$PL_{(001)}$	$p112$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$PL_{(010)}$	$pb'11$ 12.3.59	$\{E, M_y\mathcal{T}\}$	0	v
			$PL_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
13 24.5.140 $p^1m^{-1}a^{-1}2$ SI p. 85	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pma2$ 24.1.136	$\{E, C_{2z}, M_y, M_x\}$	0	0
			$AX_{[010]}$	$pm'a'2$ 24.3.138	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$AX_{[001]}$	$pm'a2'$ 24.4.139	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$PL_{(001)}$	$p112$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$PL_{(010)}$	$pb11$ 12.1.57	$\{E, M_y\}$	v	0
			$PL_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
14 25.4.146 $p^1b^{-1}a^{-1}2$ SI p. 86	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pba2$ 25.1.143	$\{E, C_{2z}, M_y, M_x\}$	0	0
			$AX_{[010]}$	$pb'a'2$ 25.3.145	$\{E, C_{2z}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$AX_{[001]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$PL_{(001)}$	$p112$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$PL_{(010)}$	$pb11$ 12.1.57	$\{E, M_y\}$	v	0
			$PL_{(100)}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
15 26.4.150 $c^1m^{-1}m^{-1}2$ SI p. 86	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$cmm2$ 26.1.147	$\{E,\, C_{2z},\, M_y,\, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$cm'm'2$ 26.3.149	$\{E,\, C_{2z},\, M_y\mathcal{T},\, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$cmm'2'$ 26.4.150	$\{E,\, C_{2z}\mathcal{T},\, M_y,\, M_x\mathcal{T}\}$	v	0
			$\mathrm{PL}_{(001)}$	$p112$ 3.1.8	$\{E,\, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$cm11$ 13.1.61	$\{E,\, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$cm'11$ 13.3.63	$\{E,\, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
16 27.3.156 $p^{-1}m^12^{-1}m$ SI p. 86	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pm'2'm$ 27.5.158	$\{E,\, C_{2y}\mathcal{T},\, M_x\mathcal{T},\, M_z\}$	+	0
			$\mathrm{AX}_{[010]}$	$pm2m$ 27.1.154	$\{E,\, C_{2y},\, M_x,\, M_z\}$	0	+
			$\mathrm{AX}_{[001]}$	$pm2'm'$ 27.4.157	$\{E,\, C_{2y}\mathcal{T},\, M_x,\, M_z\mathcal{T}\}$	0	−
			$\mathrm{PL}_{(001)}$	$p11m$ 4.1.12	$\{E,\, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E,\, C_{2y}\mathcal{T}\}$	+	−
			$\mathrm{PL}_{(100)}$	$pm11$ 11.1.50	$\{E,\, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
17 27.4.157 $p^1m^{-1}2^{-1}m$ SI p. 86	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pm2m$ 27.1.154	$\{E,\, C_{2y},\, M_x,\, M_z\}$	0	+
			$\mathrm{AX}_{[010]}$	$pm'2'm$ 27.5.158	$\{E,\, C_{2y}\mathcal{T},\, M_x\mathcal{T},\, M_z\}$	+	0
			$\mathrm{AX}_{[001]}$	$pm'2m'$ 27.3.156	$\{E,\, C_{2y},\, M_x\mathcal{T},\, M_z\mathcal{T}\}$	−	0
			$\mathrm{PL}_{(001)}$	$p11m$ 4.1.12	$\{E,\, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p211$ 8.1.34	$\{E,\, C_{2y}\}$	−	+
			$\mathrm{PL}_{(100)}$	$pm'11$ 11.3.52	$\{E,\, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
18 28.3.169 $p^{-1}m^12_1^{-1}b$ SI p. 86	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'2'_1b$ 28.4.170	$\{E,\, C_{2y}\mathcal{T},\, M_z,\, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pm2_1b$ 28.1.167	$\{E,\, C_{2y},\, M_z,\, M_x\}$	0	+
			$\mathrm{AX}_{[001]}$	$pm2'_1b'$ 28.5.171	$\{E,\, C_{2y}\mathcal{T},\, M_z\mathcal{T},\, M_x\}$	0	−
			$\mathrm{PL}_{(001)}$	$p11a$ 5.1.17	$\{E,\, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'_111$ 9.3.43	$\{E,\, C_{2y}\mathcal{T}\}$	+	−
			$\mathrm{PL}_{(100)}$	$pm11$ 11.1.50	$\{E,\, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
19 28.5.171 $p^1m^{-1}2_1^{-1}b$ SI p. 86	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm2_1b$ 28.1.167	$\{E,\ C_{2y},\ M_z,\ M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pm'2'_1b$ 28.4.170	$\{E,\ C_{2y}\mathcal{T},\ M_z,\ M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[001]}$	$pm'2_1b'$ 28.3.169	$\{E,\ C_{2y},\ M_z\mathcal{T},\ M_x\mathcal{T}\}$	−	0
			$\mathrm{PL}_{(001)}$	$p11a$ 5.1.17	$\{E,\ M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_111$ 9.1.41	$\{E,\ C_{2y}\}$	−	+
			$\mathrm{PL}_{(100)}$	$pm'11$ 11.3.52	$\{E,\ M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
20 29.3.176 $p^{-1}b^12_1^{-1}m$ SI p. 86	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pb'2'_1m$ 29.5.178	$\{E,\ C_{2y}\mathcal{T},\ M_x\mathcal{T},\ M_z\}$	+	0
			$\mathrm{AX}_{[010]}$	$pb2_1m$ 29.1.174	$\{E,\ C_{2y},\ M_x,\ M_z\}$	0	+
			$\mathrm{AX}_{[001]}$	$pb2'_1m'$ 29.4.177	$\{E,\ C_{2y}\mathcal{T},\ M_x,\ M_z\mathcal{T}\}$	0	−
			$\mathrm{PL}_{(001)}$	$p11m$ 4.1.12	$\{E,\ M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'_111$ 9.3.43	$\{E,\ C_{2y}\mathcal{T}\}$	+	−
			$\mathrm{PL}_{(100)}$	$pb11$ 12.1.57	$\{E,\ M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
21 29.4.177 $p^1b^{-1}2_1^{-1}m$ SI p. 86	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pb2_1m$ 29.1.174	$\{E,\ C_{2y},\ M_x,\ M_z\}$	0	+
			$\mathrm{AX}_{[010]}$	$pb'2'_1m$ 29.5.178	$\{E,\ C_{2y}\mathcal{T},\ M_x\mathcal{T},\ M_z\}$	+	0
			$\mathrm{AX}_{[001]}$	$pb'2_1m'$ 29.3.176	$\{E,\ C_{2y},\ M_x\mathcal{T},\ M_z\mathcal{T}\}$	−	0
			$\mathrm{PL}_{(001)}$	$p11m$ 4.1.12	$\{E,\ M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_111$ 9.1.41	$\{E,\ C_{2y}\}$	−	+
			$\mathrm{PL}_{(100)}$	$pb'11$ 12.3.59	$\{E,\ M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
22 30.3.183 $p^{-1}b^12^{-1}b$ SI p. 86	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pb'2'b$ 30.4.184	$\{E,\ C_{2y}\mathcal{T},\ M_z,\ M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pb2b$ 30.1.181	$\{E,\ C_{2y},\ M_z,\ M_x\}$	0	+
			$\mathrm{AX}_{[001]}$	$pb2'b'$ 30.5.185	$\{E,\ C_{2y}\mathcal{T},\ M_z\mathcal{T},\ M_x\}$	0	−
			$\mathrm{PL}_{(001)}$	$p11a$ 5.1.17	$\{E,\ M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E,\ C_{2y}\mathcal{T}\}$	+	−
			$\mathrm{PL}_{(100)}$	$pb11$ 12.1.57	$\{E,\ M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
23 30.5.185 $p^1b^{-1}2^{-1}b$ SI p. 86	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pb2b$ 30.1.181	$\{E, C_{2y}, M_z, M_x\}$	0	+
			$AX_{[010]}$	$pb'2'b$ 30.4.184	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}\}$	+	0
			$AX_{[001]}$	$pb'2b'$ 30.3.183	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	−	0
			$PL_{(001)}$	$p11a$ 5.1.17	$\{E, M_z\}$	+	+
			$PL_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	−	+
			$PL_{(100)}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
24 31.3.190 $p^{-1}m^12^{-1}a$ SI p. 87	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pm'2'a$ 31.4.191	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}\}$	+	0
			$AX_{[010]}$	$pm2a$ 31.1.188	$\{E, C_{2y}, M_z, M_x\}$	0	+
			$AX_{[001]}$	$pm2'a'$ 31.5.192	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	−
			$PL_{(001)}$	$p11a$ 5.1.17	$\{E, M_z\}$	+	+
			$PL_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	+	−
			$PL_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
25 (S1) 31.5.192 $p^1m^{-1}2^{-1}a$ SI p. 87	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pm2a$ 31.1.188	$\{E, C_{2y}, M_z, M_x\}$	0	+
			$AX_{[010]}$	$pm'2'a$ 31.4.191	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}\}$	+	0
			$AX_{[001]}$	$pm'2a'$ 31.3.190	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	−	0
			$PL_{(001)}$	$p11a$ 5.1.17	$\{E, M_z\}$	+	+
			$PL_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	−	+
			$PL_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
26 32.3.199 $p^{-1}m^12_1^{-1}n$ SI p. 87	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pm'2'_1n$ 32.4.200	$\{E, C_{2y}\mathcal{T}, M_z, M_x\mathcal{T}\}$	+	0
			$AX_{[010]}$	$pm2_1n$ 32.1.197	$\{E, C_{2y}, M_z, M_x\}$	0	+
			$AX_{[001]}$	$pm2'_1n'$ 32.5.201	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	−
			$PL_{(001)}$	$p11a$ 5.1.17	$\{E, M_z\}$	+	+
			$PL_{(010)}$	$p2'_111$ 9.3.43	$\{E, C_{2y}\mathcal{T}\}$	+	−
			$PL_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
27 32.5.201 $p^1m^{-1}2_1^{-1}n$ SI p. 87	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm2_1n$ 32.1.197	$\{E,\ C_{2y},\ M_z,\ M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pm'2'_1n$ 32.4.200	$\{E,\ C_{2y}\mathcal{T},\ M_z,\ M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[001]}$	$pm'2_1n'$ 32.3.199	$\{E,\ C_{2y},\ M_z\mathcal{T},\ M_x\mathcal{T}\}$	−	0
			$\mathrm{PL}_{(001)}$	$p11a$ 5.1.17	$\{E,\ M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_111$ 9.1.41	$\{E,\ C_{2y}\}$	−	+
			$\mathrm{PL}_{(100)}$	$pm'11$ 11.3.52	$\{E,\ M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
28 33.3.204 $p^{-1}b^12_1^{-1}a$ SI p. 87	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pb'2'_1a$ 33.4.205	$\{E,\ C_{2y}\mathcal{T},\ M_z,\ M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pb2_1a$ 33.1.202	$\{E,\ C_{2y},\ M_z,\ M_x\}$	0	+
			$\mathrm{AX}_{[001]}$	$pb2'_1a'$ 33.5.206	$\{E,\ C_{2y}\mathcal{T},\ M_z\mathcal{T},\ M_x\}$	0	−
			$\mathrm{PL}_{(001)}$	$p11a$ 5.1.17	$\{E,\ M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'_111$ 9.3.43	$\{E,\ C_{2y}\mathcal{T}\}$	+	−
			$\mathrm{PL}_{(100)}$	$pb11$ 12.1.57	$\{E,\ M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
29 33.5.206 $p^1b^{-1}2_1^{-1}a$ SI p. 87	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pb2_1a$ 33.1.202	$\{E,\ C_{2y},\ M_z,\ M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pb'2'_1a$ 33.4.205	$\{E,\ C_{2y}\mathcal{T},\ M_z,\ M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[001]}$	$pb'2_1a'$ 33.3.204	$\{E,\ C_{2y},\ M_z\mathcal{T},\ M_x\mathcal{T}\}$	−	0
			$\mathrm{PL}_{(001)}$	$p11a$ 5.1.17	$\{E,\ M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_111$ 9.1.41	$\{E,\ C_{2y}\}$	−	+
			$\mathrm{PL}_{(100)}$	$pb'11$ 12.3.59	$\{E,\ M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
30 34.3.209 $p^{-1}b^12^{-1}n$ SI p. 87	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pb'2'n$ 34.5.211	$\{E,\ C_{2y}\mathcal{T},\ M_x\mathcal{T},\ M_z\}$	+	0
			$\mathrm{AX}_{[010]}$	$pb2n$ 34.1.207	$\{E,\ C_{2y},\ M_x,\ M_z\}$	0	+
			$\mathrm{AX}_{[001]}$	$pb2'n'$ 34.4.210	$\{E,\ C_{2y}\mathcal{T},\ M_x,\ M_z\mathcal{T}\}$	0	−
			$\mathrm{PL}_{(001)}$	$p11a$ 5.1.17	$\{E,\ M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E,\ C_{2y}\mathcal{T}\}$	+	−
			$\mathrm{PL}_{(100)}$	$pb11$ 12.1.57	$\{E,\ M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
31 34.4.210 $p^1b^{-1}2^{-1}n$ SI p. 87	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$	$AX_{[100]}$	$pb2n$ 34.1.207	$\{E, C_{2y}, M_x, M_z\}$	0	+
			$AX_{[010]}$	$pb'2'n$ 34.5.211	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\}$	+	0
			$AX_{[001]}$	$pb'2n'$ 34.3.209	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	−	0
			$PL_{(001)}$	$p11a$ 5.1.17	$\{E, M_z\}$	+	+
			$PL_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	−	+
			$PL_{(100)}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
32 35.3.214 $c^{-1}m^12^{-1}m$ SI p. 87	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$AX_{[100]}$	$cm'2'm$ 35.5.216	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\}$	+	0
			$AX_{[010]}$	$cm2m$ 35.1.212	$\{E, C_{2y}, M_x, M_z\}$	0	+
			$AX_{[001]}$	$cm2'm'$ 35.4.215	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	−
			$PL_{(001)}$	$p11m$ 4.1.12	$\{E, M_z\}$	+	+
			$PL_{(010)}$	$c2'11$ 10.3.47	$\{E, C_{2y}\mathcal{T}\}$	+	−
			$PL_{(100)}$	$cm11$ 13.1.61	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
33 35.4.215 $c^1m^{-1}2^{-1}m$ SI p. 87	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$	$AX_{[100]}$	$cm2m$ 35.1.212	$\{E, C_{2y}, M_x, M_z\}$	0	+
			$AX_{[010]}$	$cm'2'm$ 35.5.216	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\}$	+	0
			$AX_{[001]}$	$cm'2m'$ 35.3.214	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	−	0
			$PL_{(001)}$	$p11m$ 4.1.12	$\{E, M_z\}$	+	+
			$PL_{(010)}$	$c211$ 10.1.45	$\{E, C_{2y}\}$	−	+
			$PL_{(100)}$	$cm'11$ 13.3.63	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
34 36.3.223 $c^{-1}m^12^{-1}a$ SI p. 88	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$AX_{[100]}$	$cm'2'a$ 36.5.225	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\}$	+	0
			$AX_{[010]}$	$cm2a$ 36.1.221	$\{E, C_{2y}, M_x, M_z\}$	0	+
			$AX_{[001]}$	$cm2'a'$ 36.4.224	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	−
			$PL_{(001)}$	$p11a$ 5.1.17	$\{E, M_z\}$	+	+
			$PL_{(010)}$	$c2'11$ 10.3.47	$\{E, C_{2y}\mathcal{T}\}$	+	−
			$PL_{(100)}$	$cm11$ 13.1.61	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
35 36.4.224 $c^1m^{-1}2^{-1}a$ SI p. 88	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$cm2a$ 36.1.221	$\{E, C_{2y}, M_x, M_z\}$	0	+
			$\mathrm{AX}_{[010]}$	$cm'2'a$ 36.5.225	$\{E, C_{2y}\mathcal{T}, M_x\mathcal{T}, M_z\}$	+	0
			$\mathrm{AX}_{[001]}$	$cm'2a'$ 36.3.223	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	−	0
			$\mathrm{PL}_{(001)}$	$p11a$ 5.1.17	$\{E, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$c211$ 10.1.45	$\{E, C_{2y}\}$	−	+
			$\mathrm{PL}_{(100)}$	$cm'11$ 13.3.63	$\{E, M_x\mathcal{T}\}$	v	0
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
36 37.5.234 $p^1m^{-1}m^{-1}m$ SI p. 88	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmmm$ 37.1.230	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pm'm'm$ 37.4.233	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	−	0
			$\mathrm{PL}_{(001)}$	$p112/m$ 6.1.21	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2/m11$ 14.1.66	$\{E, C_{2y}, \mathcal{P}, M_y\}$	−	0
			$\mathrm{PL}_{(100)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	−	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	−	−
37 38.4.246 $p^1m^{-1}a^{-1}a$ SI p. 88	Type-IV	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pmaa$ 38.1.243	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}, M_x, M_z, M_y\}$	0	0
			$\mathrm{AX}_{[010]}$	$pm'a'a$ 38.5.247	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, M_z, M_y\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pm'aa'$ 38.6.248	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\}$	−	0
			$\mathrm{PL}_{(001)}$	$p112/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2/b11$ 16.1.83	$\{E, C_{2y}, \mathcal{P}, M_y\}$	−	0
			$\mathrm{PL}_{(100)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	−	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	−	−
38 38.6.248 $p^{-1}m^1a^{-1}a$ SI p. 88	Type-IV	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pm'a'a$ 38.5.247	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, M_z, M_y\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$pmaa$ 38.1.243	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}, M_x, M_z, M_y\}$	0	0
			$\mathrm{AX}_{[001]}$	$pma'a'$ 38.4.246	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_x, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	−
			$\mathrm{PL}_{(001)}$	$p112/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	−
			$\mathrm{PL}_{(100)}$	$p2/m11$ 14.1.66	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	−
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	−	−

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
39 (S2) 39.5.260 $p^1b^{-1}a^{-1}n$ SI p. 89	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pban$ 39.1.256	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pb'a'n$ 39.4.259	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}_{(001)}$	$p112/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2/b11$ 16.1.83	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
40 40.4.266 $p^{-1}m^1a^{-1}m$ SI p. 89	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'a'm$ 40.5.267	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$pmam$ 40.1.263	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}, M_y, M_z, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$pma'm'$ 40.6.268	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$\mathrm{PL}_{(001)}$	$p112/m$ 6.1.21	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
41 40.6.268 $p^1m^{-1}a^{-1}m$ SI p. 89	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmam$ 40.1.263	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}, M_y, M_z, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pm'a'm$ 40.5.267	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_z, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pm'am'$ 40.4.266	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_y, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}_{(001)}$	$p112/m$ 6.1.21	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2/b11$ 16.1.83	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
42 41.5.280 $p^{-1}m^1m^{-1}a$ SI p. 89	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'm'a$ 41.4.279	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$pmma$ 41.1.276	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$pmm'a'$ 41.6.281	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{PL}_{(001)}$	$p112/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
43 41.6.281 $p^1m^{-1}m^{-1}a$ SI p. 89	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pmma$ 41.1.276	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$AX_{[010]}$	$pm'm'a$ 41.4.279	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$AX_{[001]}$	$pm'ma'$ 41.5.280	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$PL_{(001)}$	$p11\bar{2}/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$PL_{(010)}$	$p2/m11$ 14.1.66	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$PL_{(100)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
44 42.4.292 $p^{-1}m^1a^{-1}n$ SI p. 89	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pm'a'n$ 42.5.293	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_z, M_x\mathcal{T}\}$	0	0
			$AX_{[010]}$	$pman$ 42.1.289	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}, M_y, M_z, M_x\}$	0	0
			$AX_{[001]}$	$pma'n'$ 42.6.294	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$PL_{(001)}$	$p11\bar{2}/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$PL_{(010)}$	$p2'_1/b'11$ 17.4.93	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$PL_{(100)}$	$p2/m11$ 14.1.66	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
45 42.6.294 $p^1m^{-1}a^{-1}n$ SI p. 90	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pman$ 42.1.289	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}, M_y, M_z, M_x\}$	0	0
			$AX_{[010]}$	$pm'a'n$ 42.5.293	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_z, M_x\mathcal{T}\}$	0	0
			$AX_{[001]}$	$pm'an'$ 42.4.292	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_y, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$PL_{(001)}$	$p11\bar{2}/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$PL_{(010)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$PL_{(100)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
46 43.4.301 $p^1b^{-1}a^{-1}a$ SI p. 90	Type-IV	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$	$AX_{[100]}$	$pbaa$ 43.1.298	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}, M_x, M_z, M_y\}$	0	0
			$AX_{[010]}$	$pb'a'a$ 43.6.303	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, M_z, M_y\mathcal{T}\}$	0	0
			$AX_{[001]}$	$pb'aa'$ 43.5.302	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\}$	–	0
			$PL_{(001)}$	$p11\bar{2}/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$PL_{(010)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$PL_{(100)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
47 43.5.302 $p^{-1}b^{1}a^{-1}a$ SI p. 90	Type-IV	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pb'a'a$ 43.6.303	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}, M_z, M_y\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$pbaa$ 43.1.298	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}, M_x, M_z, M_y\}$	0	0
			$\mathrm{AX}_{[001]}$	$pba'a'$ 43.4.301	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_x, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(001)}$	$p11\bar{2}/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'_1/b'11$ 17.4.93	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2/b11$ 16.1.83	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
48 44.5.311 $p^{1}b^{-1}a^{-1}m$ SI p. 90	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pbam$ 44.1.307	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pb'a'm$ 44.4.310	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}_{(001)}$	$p11\bar{2}/m$ 6.1.21	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2'_1/b'11$ 17.4.93	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
49 45.4.317 $p^{-1}b^{1}m^{-1}a$ SI p. 90	Type-IV	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pb'm'a$ 45.6.319	$\{E, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2z}, \mathcal{P}, M_y\mathcal{T}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$pbma$ 45.1.314	$\{E, C_{2y}, C_{2x}, C_{2z}, \mathcal{P}, M_y, M_x, M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	$pbm'a'$ 45.5.318	$\{E, C_{2y}\mathcal{T}, C_{2x}, C_{2z}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(001)}$	$p11\bar{2}/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
50 45.5.318 $p^{1}b^{-1}m^{-1}a$ SI p. 91	Type-IV	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pbma$ 45.1.314	$\{E, C_{2y}, C_{2x}, C_{2z}, \mathcal{P}, M_y, M_x, M_z\}$	0	0
			$\mathrm{AX}_{[010]}$	$pb'm'a$ 45.6.319	$\{E, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2z}, \mathcal{P}, M_y\mathcal{T}, M_x\mathcal{T}, M_z\}$	0	0
			$\mathrm{AX}_{[001]}$	$pb'ma'$ 45.4.317	$\{E, C_{2y}, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}, M_y, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$\mathrm{PL}_{(001)}$	$p11\bar{2}/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2'_1/b'11$ 17.4.93	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
51 46.5.327 $p^1m^{-1}m^{-1}n$ SI p. 91	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmmn$ 46.1.323	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pm'm'n$ 46.4.326	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}_{(001)}$	$p\bar{1}12/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2'_1/m'11$ 15.4.79	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
52 47.5.334 $c^1m^{-1}m^{-1}m$ SI p. 91	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$cmmm$ 47.1.330	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$cm'm'm$ 47.4.333	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}_{(001)}$	$p\bar{1}12/m$ 6.1.21	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2/m11$ 18.1.95	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
53 48.5.347 $c^1m^{-1}m^{-1}e$ SI p. 91	Type-IV	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$cmme$ 48.1.343	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$cm'm'e$ 48.4.346	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_z, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}_{(001)}$	$p\bar{1}12/a$ 7.1.28	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2/m11$ 18.1.95	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
54 73.3.491 $p^{-1}6$ SI p. 92	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{6z}^{-1}\}$ $\{-1\|C_{6z}\}$	$\mathrm{AX}_{[001]}$	$p6'$ 73.3.491	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}\mathcal{T}, C_{6z}^{-1}\mathcal{T}, C_{6z}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112$ 3.1.8	$\{E, C_{2z}\}$	0	0
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
55 74.3.494 $p^{-1}\bar{6}$ SI p. 92	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$	$\mathrm{AX}_{[001]}$	$p\bar{6}'$ 74.3.494	$\{E, C_{3z}, C_{3z}^{-1}, M_z\mathcal{T}, \bar{C}_{6z}^{-1}\mathcal{T}, \bar{C}_{6z}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11m$ 4.1.12	$\{E, M_z\}$	+	+
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
56 75.4.498 $p^{-1}6/^{-1}m$ SI p. 92	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{6z}^{-1}\}$ $\{-1\|C_{6z}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$	$\mathrm{AX}_{[001]}$	$p6'/m'$ 75.4.498	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}\mathcal{T}, C_{6z}^{-1}\mathcal{T}, C_{6z}\mathcal{T}, \mathcal{P}, \bar{C}_{3z}, \bar{C}_{3z}^{-1}, M_z\mathcal{T}, \bar{C}_{6z}^{-1}\mathcal{T}, \bar{C}_{6z}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112/m$ 6.1.21	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
57 76.4.503 $p^{-1}6^12^{-1}2$ SI p. 92	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{6z}^{-1}\}$ $\{-1\|C_{6z}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$ $\{-1\|C_{2,[210]}\}$ $\{-1\|C_{2,[120]}\}$ $\{-1\|C_{2,[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$c222$ 22.1.122	$\{E, C_{2z}, C_{2,[100]}, C_{2,[120]}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c222$ 22.1.122	$\{E, C_{2z}, C_{2,[010]}, C_{2,[210]}\}$	0	0
			$\mathrm{AX}_{[110]}$	$c222$ 22.1.122	$\{E, C_{2z}, C_{2,[110]}, C_{2,[1\bar{1}0]}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c2'2'2$ 22.3.124	$\{E, C_{2z}, C_{2,[110]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[120]}$	$c2'2'2$ 22.3.124	$\{E, C_{2z}, C_{2,[100]}\mathcal{T}, C_{2,[120]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[210]}$	$c2'2'2$ 22.3.124	$\{E, C_{2z}, C_{2,[010]}\mathcal{T}, C_{2,[210]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p6'2'2$ 76.5.504	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}\mathcal{T}, C_{6z}^{-1}\mathcal{T}, C_{6z}\mathcal{T}, C_{2,[110]}\mathcal{T}, C_{2,[100]}\mathcal{T}, C_{2,[010]}\mathcal{T}, C_{2,[210]}, C_{2,[120]}, C_{2,[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p\bar{1}12$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c211$ 10.1.45	$\{E, C_{2,[120]}\}$	–	+
			$\mathrm{PL}^{*}_{(100)}$	$c211$ 10.1.45	$\{E, C_{2,[210]}\}$	v	v
			$\mathrm{PL}^{*}_{(110)}$	$c2'11$ 10.3.47	$\{E, C_{2,[110]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2,[1\bar{1}0]}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2,[100]}\mathcal{T}\}$	–	+
			$\mathrm{PL}^{*}_{(1\bar{2}0)}$	$c2'11$ 10.3.47	$\{E, C_{2,[010]}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
58 76.5.504 $p^{-1}6^{-1}2^12$ SI p. 92	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{6z}^{-1}\}$ $\{-1\|C_{6z}\}$ $\{-1\|C_{2,[110]}\}$ $\{-1\|C_{2,[100]}\}$ $\{-1\|C_{2,[010]}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$c2'2'2$ 22.3.124	$\{E, C_{2z}, C_{2,[100]}\mathcal{T}, C_{2,[120]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$c2'2'2$ 22.3.124	$\{E, C_{2z}, C_{2,[010]}\mathcal{T}, C_{2,[210]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[110]}$	$c2'2'2$ 22.3.124	$\{E, C_{2z}, C_{2,[110]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$c222$ 22.1.122	$\{E, C_{2z}, C_{2,[110]}, C_{2,[1\bar{1}0]}\}$	0	0
			$\mathrm{AX}_{[120]}$	$c222$ 22.1.122	$\{E, C_{2z}, C_{2,[100]}, C_{2,[120]}\}$	0	0
			$\mathrm{AX}_{[210]}$	$c222$ 22.1.122	$\{E, C_{2z}, C_{2,[010]}, C_{2,[210]}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p6'22'$ 76.4.503	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}\mathcal{T}, C_{6z}^{-1}\mathcal{T}, C_{6z}\mathcal{T}, C_{2,[110]}, C_{2,[100]}, C_{2,[010]}, C_{2,[210]}\mathcal{T}, C_{2,[120]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p\bar{1}12$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2'11$ 10.3.47	$\{E, C_{2,[120]}\mathcal{T}\}$	+	−
			$\mathrm{PL}^*_{(100)}$	$c2'11$ 10.3.47	$\{E, C_{2,[210]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E, C_{2,[110]}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2,[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2,[100]}\}$	+	−
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c211$ 10.1.45	$\{E, C_{2,[010]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
59 77.4.508 $p^{-1}6^1m^{-1}m$ SI p. 93	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C^{-1}_{3z}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C^{-1}_{6z}\}$ $\{-1\|C_{6z}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cmm2$ 26.1.147	$\{E, C_{2z}, M_{[100]}, M_{[120]}\}$	0	0
			$\mathrm{AX}_{[010]}$	$cmm2$ 26.1.147	$\{E, C_{2z}, M_{[010]}, M_{[210]}\}$	0	0
			$\mathrm{AX}_{[110]}$	$cmm2$ 26.1.147	$\{E, C_{2z}, M_{[110]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$cm'm'2$ 26.3.149	$\{E, C_{2z}, M_{[110]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[120]}$	$cm'm'2$ 26.3.149	$\{E, C_{2z}, M_{[100]}\mathcal{T}, M_{[120]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[210]}$	$cm'm'2$ 26.3.149	$\{E, C_{2z}, M_{[010]}\mathcal{T}, M_{[210]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p6'm'm$ 77.5.509	$\{E, C_{3z}, C^{-1}_{3z}, C_{2z}\mathcal{T}, C^{-1}_{6z}\mathcal{T}, C_{6z}\mathcal{T}, M_{[110]}\mathcal{T}, M_{[100]}\mathcal{T}, M_{[010]}\mathcal{T}, M_{[210]}, M_{[120]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p\bar{1}12$ 3.1.8	$\{E, C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$cm11$ 13.1.61	$\{E, M_{[120]}\}$	v	0
			$\mathrm{PL}^*_{(100)}$	$cm11$ 13.1.61	$\{E, M_{[210]}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E, M_{[110]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{[1\bar{1}0]}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$cm'11$ 13.3.63	$\{E, M_{[100]}\mathcal{T}\}$	v	0
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$cm'11$ 13.3.63	$\{E, M_{[010]}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
60 77.5.509 $p^{-1}6^{-1}m^1m$ SI p. 93	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{6z}^{-1}\}$ $\{-1\|C_{6z}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm'm'2$ 26.3.149	$\{E,\ C_{2z},\ M_{[100]}\mathcal{T},\ M_{[120]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$cm'm'2$ 26.3.149	$\{E,\ C_{2z},\ M_{[010]}\mathcal{T},\ M_{[210]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[110]}$	$cm'm'2$ 26.3.149	$\{E,\ C_{2z},\ M_{[110]}\mathcal{T},\ M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$cmm2$ 26.1.147	$\{E,\ C_{2z},\ M_{[110]},\ M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{AX}_{[120]}$	$cmm2$ 26.1.147	$\{E,\ C_{2z},\ M_{[100]},\ M_{[120]}\}$	0	0
			$\mathrm{AX}_{[210]}$	$cmm2$ 26.1.147	$\{E,\ C_{2z},\ M_{[010]},\ M_{[210]}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p6'mm'$ 77.4.508	$\{E,\ C_{3z},\ C_{3z}^{-1},\ C_{2z}\mathcal{T},\ C_{6z}^{-1}\mathcal{T},\ C_{6z}\mathcal{T},\ M_{[110]},\ M_{[100]},\ M_{[010]},\ M_{[210]}\mathcal{T},\ M_{[120]}\mathcal{T},\ M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p\bar{1}12$ 3.1.8	$\{E,\ C_{2z}\}$	0	0
			$\mathrm{PL}_{(010)}$	$cm'11$ 13.3.63	$\{E,\ M_{[120]}\mathcal{T}\}$	0	v
			$\mathrm{PL}^{*}_{(100)}$	$cm'11$ 13.3.63	$\{E,\ M_{[210]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^{*}_{(110)}$	$cm11$ 13.1.61	$\{E,\ M_{[110]}\}$	v	v
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E,\ M_{[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$cm11$ 13.1.61	$\{E,\ M_{[100]}\}$	0	v
			$\mathrm{PL}^{*}_{(1\bar{2}0)}$	$cm11$ 13.1.61	$\{E,\ M_{[010]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
61 78.4.513 $p^{-1}\bar{6}^1m^{-1}2$ SI p. 93	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$ $\{-1\|C_{2,[210]}\}$ $\{-1\|C_{2,[120]}\}$ $\{-1\|C_{2,[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm2m$ 35.1.212	$\{E, M_z, M_{[100]}, C_{2,[120]}\}$	0	+
			$\mathrm{AX}^*_{[010]}$	$cm2m$ 35.1.212	$\{E, M_z, M_{[010]}, C_{2,[210]}\}$	+	+
			$\mathrm{AX}^*_{[110]}$	$cm2m$ 35.1.212	$\{E, M_z, M_{[110]}, C_{2,[1\bar{1}0]}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cm'2'm$ 35.5.216	$\{E, M_z, M_{[110]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[120]}$	$cm'2'm$ 35.5.216	$\{E, M_z, M_{[100]}\mathcal{T}, C_{2,[120]}\mathcal{T}\}$	+	0
			$\mathrm{AX}^*_{[210]}$	$cm'2'm$ 35.5.216	$\{E, M_z, M_{[010]}\mathcal{T}, C_{2,[210]}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p\bar{6}'m'2$ 78.5.514	$\{E, C_{3z}, C_{3z}^{-1}, M_z\mathcal{T}, \bar{C}_{6z}^{-1}\mathcal{T}, \bar{C}_{6z}\mathcal{T}, M_{[110]}\mathcal{T}, M_{[100]}\mathcal{T}, M_{[010]}\mathcal{T}, C_{2,[210]}, C_{2,[120]}, C_{2,[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11m$ 4.1.12	$\{E, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$c211$ 10.1.45	$\{E, C_{2,[120]}\}$	−	+
			$\mathrm{PL}^*_{(100)}$	$c211$ 10.1.45	$\{E, C_{2,[210]}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E, M_{[110]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2,[1\bar{1}0]}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$cm'11$ 13.3.63	$\{E, M_{[100]}\mathcal{T}\}$	v	0
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$cm'11$ 13.3.63	$\{E, M_{[010]}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
62 (S3) 78.5.514 $p^{-1}\bar{6}^{-1}m^{1}2$ SI p. 94	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm'2'm$ 35.5.216	$\{E,\ M_z,\ M_{[100]}\mathcal{T},\ C_{2,[120]}\mathcal{T}\}$	+	0
			$\mathrm{AX}^*_{[010]}$	$cm'2'm$ 35.5.216	$\{E,\ M_z,\ M_{[010]}\mathcal{T},\ C_{2,[210]}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[110]}$	$cm'2'm$ 35.5.216	$\{E,\ M_z,\ M_{[110]}\mathcal{T},\ C_{2,[1\bar{1}0]}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cm2m$ 35.1.212	$\{E,\ M_z,\ M_{[110]},\ C_{2,[1\bar{1}0]}\}$	+	+
			$\mathrm{AX}_{[120]}$	$cm2m$ 35.1.212	$\{E,\ M_z,\ M_{[100]},\ C_{2,[120]}\}$	0	+
			$\mathrm{AX}^*_{[210]}$	$cm2m$ 35.1.212	$\{E,\ M_z,\ M_{[010]},\ C_{2,[210]}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p\bar{6}'m2'$ 78.4.513	$\{E,\ C_{3z},\ C_{3z}^{-1},\ M_z\mathcal{T},\ \bar{C}_{6z}^{-1}\mathcal{T},\ \bar{C}_{6z}\mathcal{T},\ M_{[110]},\ M_{[100]},\ M_{[010]},\ C_{2,[210]}\mathcal{T},\ C_{2,[120]}\mathcal{T},\ C_{2,[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11m$ 4.1.12	$\{E,\ M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[120]}\mathcal{T}\}$	+	−
			$\mathrm{PL}^*_{(100)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[210]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E,\ M_{[110]}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E,\ C_{2,[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$cm11$ 13.1.61	$\{E,\ M_{[100]}\}$	0	v
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$cm11$ 13.1.61	$\{E,\ M_{[010]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
63 79.4.518 $p^{-1}\bar{6}^{1}2^{-1}m$ SI p. 94	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm2m$ 35.1.212	$\{E, M_z, C_{2,[100]}, M_{[120]}\}$	+	0
			$\mathrm{AX}^{*}_{[010]}$	$cm2m$ 35.1.212	$\{E, M_z, C_{2,[010]}, M_{[210]}\}$	+	+
			$\mathrm{AX}^{*}_{[110]}$	$cm2m$ 35.1.212	$\{E, M_z, C_{2,[110]}, M_{[1\bar{1}0]}\}$	+	+
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$cm'2'm$ 35.5.216	$\{E, M_z, C_{2,[110]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[120]}$	$cm'2'm$ 35.5.216	$\{E, M_z, C_{2,[100]}\mathcal{T}, M_{[120]}\mathcal{T}\}$	0	+
			$\mathrm{AX}^{*}_{[210]}$	$cm'2'm$ 35.5.216	$\{E, M_z, C_{2,[010]}\mathcal{T}, M_{[210]}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p\bar{6}'2'm$ 79.5.519	$\{E, C_{3z}, C_{3z}^{-1}, M_z\mathcal{T}, \bar{C}_{6z}^{-1}\mathcal{T}, \bar{C}_{6z}\mathcal{T}, C_{2,[110]}\mathcal{T}, C_{2,[100]}\mathcal{T}, C_{2,[010]}\mathcal{T}, M_{[210]}, M_{[120]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11m$ 4.1.12	$\{E, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$cm11$ 13.1.61	$\{E, M_{[120]}\}$	v	0
			$\mathrm{PL}^{*}_{(100)}$	$cm11$ 13.1.61	$\{E, M_{[210]}\}$	v	v
			$\mathrm{PL}^{*}_{(110)}$	$c2'11$ 10.3.47	$\{E, C_{2,[110]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{[1\bar{1}0]}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2,[100]}\mathcal{T}\}$	−	+
			$\mathrm{PL}^{*}_{(1\bar{2}0)}$	$c2'11$ 10.3.47	$\{E, C_{2,[010]}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
64 79.5.519 $p^{-1}\bar{6}^{-1}2^1m$ SI p. 94	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$ $\{-1\|C_{2,[110]}\}$ $\{-1\|C_{2,[100]}\}$ $\{-1\|C_{2,[010]}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm'2'm$ 35.5.216	$\{E,\ M_z,\ C_{2,[100]}\mathcal{T},\ M_{[120]}\mathcal{T}\}$	0	+
			$\mathrm{AX}^*_{[010]}$	$cm'2'm$ 35.5.216	$\{E,\ M_z,\ C_{2,[010]}\mathcal{T},\ M_{[210]}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[110]}$	$cm'2'm$ 35.5.216	$\{E,\ M_z,\ C_{2,[110]}\mathcal{T},\ M_{[1\bar{1}0]}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cm2m$ 35.1.212	$\{E,\ M_z,\ C_{2,[110]},\ M_{[1\bar{1}0]}\}$	+	+
			$\mathrm{AX}_{[120]}$	$cm2m$ 35.1.212	$\{E,\ M_z,\ C_{2,[100]},\ M_{[120]}\}$	+	0
			$\mathrm{AX}^*_{[210]}$	$cm2m$ 35.1.212	$\{E,\ M_z,\ C_{2,[010]},\ M_{[210]}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p\bar{6}'2m'$ 79.4.518	$\{E,\ C_{3z},\ C_{3z}^{-1},\ M_z\mathcal{T},\ \bar{C}_{6z}^{-1}\mathcal{T},\ \bar{C}_{6z}\mathcal{T},\ C_{2,[110]},\ C_{2,[100]},\ C_{2,[010]},\ M_{[210]}\mathcal{T},\ M_{[120]}\mathcal{T},\ M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11m$ 4.1.12	$\{E,\ M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$cm'11$ 13.3.63	$\{E,\ M_{[120]}\mathcal{T}\}$	0	v
			$\mathrm{PL}^*_{(100)}$	$cm'11$ 13.3.63	$\{E,\ M_{[210]}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E,\ C_{2,[110]}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E,\ M_{[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$c211$ 10.1.45	$\{E,\ C_{2,[100]}\}$	+	−
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c211$ 10.1.45	$\{E,\ C_{2,[010]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
65 80.6.525 $p^{-1}6/^{-1}m^1m^{-1}m$ SI p. 94	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{6z}^{-1}\}$ $\{-1\|C_{6z}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$ $\{-1\|C_{2,[210]}\}$ $\{-1\|C_{2,[120]}\}$ $\{-1\|C_{2,[1\bar{1}0]}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cmmm$ 47.1.330	$\{E, C_{2z}, C_{2,[100]}, C_{2,[120]}, \mathcal{P}, M_z, M_{[100]}, M_{[120]}\}$	0	0
			$\mathrm{AX}_{[010]}$	$cmmm$ 47.1.330	$\{E, C_{2z}, C_{2,[010]}, C_{2,[210]}, \mathcal{P}, M_z, M_{[010]}, M_{[210]}\}$	0	0
			$\mathrm{AX}_{[110]}$	$cmmm$ 47.1.330	$\{E, C_{2z}, C_{2,[110]}, C_{2,[1\bar{1}0]}, \mathcal{P}, M_z, M_{[110]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$cm'm'm$ 47.4.333	$\{E, C_{2z}, C_{2,[110]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}, M_z, M_{[110]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[120]}$	$cm'm'm$ 47.4.333	$\{E, C_{2z}, C_{2,[100]}\mathcal{T}, C_{2,[120]}\mathcal{T}, \mathcal{P}, M_z, M_{[100]}\mathcal{T}, M_{[120]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[210]}$	$cm'm'm$ 47.4.333	$\{E, C_{2z}, C_{2,[010]}\mathcal{T}, C_{2,[210]}\mathcal{T}, \mathcal{P}, M_z, M_{[010]}\mathcal{T}, M_{[210]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p6'/m'm'm$ 80.7.526	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}\mathcal{T}, C_{6z}^{-1}\mathcal{T}, C_{6z}\mathcal{T}, C_{2,[110]}\mathcal{T}, C_{2,[100]}\mathcal{T}, C_{2,[010]}\mathcal{T}, C_{2,[210]}, C_{2,[120]}, C_{2,[1\bar{1}0]}, \mathcal{P}, \bar{C}_{3z}, \bar{C}_{3z}^{-1}, M_z\mathcal{T}, \bar{C}_{6z}^{-1}\mathcal{T}, \bar{C}_{6z}\mathcal{T}, M_{[110]}\mathcal{T}, M_{[100]}\mathcal{T}, M_{[010]}\mathcal{T}, M_{[210]}, M_{[120]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11\bar{2}/m$ 6.1.21	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[120]}, \mathcal{P}, M_{[120]}\}$	–	0
			$\mathrm{PL}^*_{(100)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[210]}, \mathcal{P}, M_{[210]}\}$	–	–
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[110]}\mathcal{T}, \mathcal{P}, M_{[110]}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[1\bar{1}0]}, \mathcal{P}, M_{[1\bar{1}0]}\}$	–	–
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[100]}\mathcal{T}, \mathcal{P}, M_{[100]}\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[010]}\mathcal{T}, \mathcal{P}, M_{[010]}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
66 80.7.526 $p^{-1}6/^{-1}m^{-1}m^1m$ SI p. 95	Type-IV	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{6z}^{-1}\}$ $\{-1\|C_{6z}\}$ $\{-1\|C_{2,[110]}\}$ $\{-1\|C_{2,[100]}\}$ $\{-1\|C_{2,[010]}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm'm'm$ 47.4.333	$\{E, C_{2z}, C_{2,[100]}\mathcal{T}, C_{2,[120]}\mathcal{T}, \mathcal{P}, M_z, M_{[100]}\mathcal{T}, M_{[120]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$cm'm'm$ 47.4.333	$\{E, C_{2z}, C_{2,[010]}\mathcal{T}, C_{2,[210]}\mathcal{T}, \mathcal{P}, M_z, M_{[010]}\mathcal{T}, M_{[210]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[110]}$	$cm'm'm$ 47.4.333	$\{E, C_{2z}, C_{2,[110]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}, M_z, M_{[110]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$cmmm$ 47.1.330	$\{E, C_{2z}, C_{2,[110]}, C_{2,[1\bar{1}0]}, \mathcal{P}, M_z, M_{[110]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{AX}_{[120]}$	$cmmm$ 47.1.330	$\{E, C_{2z}, C_{2,[100]}, C_{2,[120]}, \mathcal{P}, M_z, M_{[100]}, M_{[120]}\}$	0	0
			$\mathrm{AX}_{[210]}$	$cmmm$ 47.1.330	$\{E, C_{2z}, C_{2,[010]}, C_{2,[210]}, \mathcal{P}, M_z, M_{[010]}, M_{[210]}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p6'/m'mm'$ 80.6.525	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}\mathcal{T}, C_{6z}^{-1}\mathcal{T}, C_{6z}\mathcal{T}, C_{2,[110]}, C_{2,[100]}, C_{2,[010]}, C_{2,[210]}\mathcal{T}, C_{2,[120]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}, \bar{C}_{3z}, \bar{C}_{3z}^{-1}, M_z\mathcal{T}, \bar{C}_{6z}^{-1}\mathcal{T}, \bar{C}_{6z}\mathcal{T}, M_{[110]}, M_{[100]}, M_{[010]}, M_{[210]}\mathcal{T}, M_{[120]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p1\bar{1}2/m$ 6.1.21	$\{E, C_{2z}, \mathcal{P}, M_z\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[120]}\mathcal{T}, \mathcal{P}, M_{[120]}\mathcal{T}\}$	0	–
			$\mathrm{PL}^*_{(100)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[210]}\mathcal{T}, \mathcal{P}, M_{[210]}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(110)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[110]}, \mathcal{P}, M_{[110]}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}, M_{[1\bar{1}0]}\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[100]}, \mathcal{P}, M_{[100]}\}$	0	–
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[010]}, \mathcal{P}, M_{[010]}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

Appendix A. Complete coupled-component constraints

$A^\ell = \chi^\ell_{xyy}$, $B^\ell = \chi^\ell_{yxx}$. Each row below gives the entire allowed linear space for a starred branch, with independent real parameters u or u_1, u_2. These relations supplement the $+, -, 0, v$ flags; they are not additional assumptions.

No.	cSLG label	Néel direction	Complete allowed space
57	76.4.503	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
57	76.4.503	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
57	76.4.503	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
57	76.4.503	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
58	76.5.504	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
58	76.5.504	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
58	76.5.504	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
58	76.5.504	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
59	77.4.508	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
59	77.4.508	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
59	77.4.508	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
59	77.4.508	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
60	77.5.509	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
60	77.5.509	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
60	77.5.509	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
60	77.5.509	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
61	78.4.513	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
61	78.4.513	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
61	78.4.513	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
61	78.4.513	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
61	78.4.513	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
61	78.4.513	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
61	78.4.513	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
61	78.4.513	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
62	78.5.514	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
62	78.5.514	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
62	78.5.514	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
62	78.5.514	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
62	78.5.514	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
62	78.5.514	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
62	78.5.514	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
62	78.5.514	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
63	79.4.518	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
63	79.4.518	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
63	79.4.518	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
63	79.4.518	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
63	79.4.518	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
63	79.4.518	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
63	79.4.518	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
63	79.4.518	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
64	79.5.519	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
64	79.5.519	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
64	79.5.519	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
64	79.5.519	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
64	79.5.519	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
64	79.5.519	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$

No.	cSLG label	Néel direction	Complete allowed space
64	79.5.519	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\, 1,\, 0,\, 0\big) + u_2\big(0,\, 0,\, -\sqrt{3},\, 1\big)$
64	79.5.519	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\, -\sqrt{3},\, 2,\, 0\big) + u_2\big(-\sqrt{3},\, 1,\, 0,\, 2\big)$
65	80.6.525	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -3,\, -\sqrt{3},\, 3\big)$
65	80.6.525	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -3,\, \sqrt{3},\, 3\big)$
65	80.6.525	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -3,\, \sqrt{3},\, 3\big)$
65	80.6.525	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -3,\, -\sqrt{3},\, 3\big)$
66	80.7.526	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -1,\, \sqrt{3},\, 1\big)$
66	80.7.526	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -1,\, -\sqrt{3},\, 1\big)$
66	80.7.526	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -1,\, -\sqrt{3},\, 1\big)$
66	80.7.526	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -1,\, \sqrt{3},\, 1\big)$